

Higher-order clustering for 3D point clouds

Exact density hierarchies and geometric tokens for 3D learning

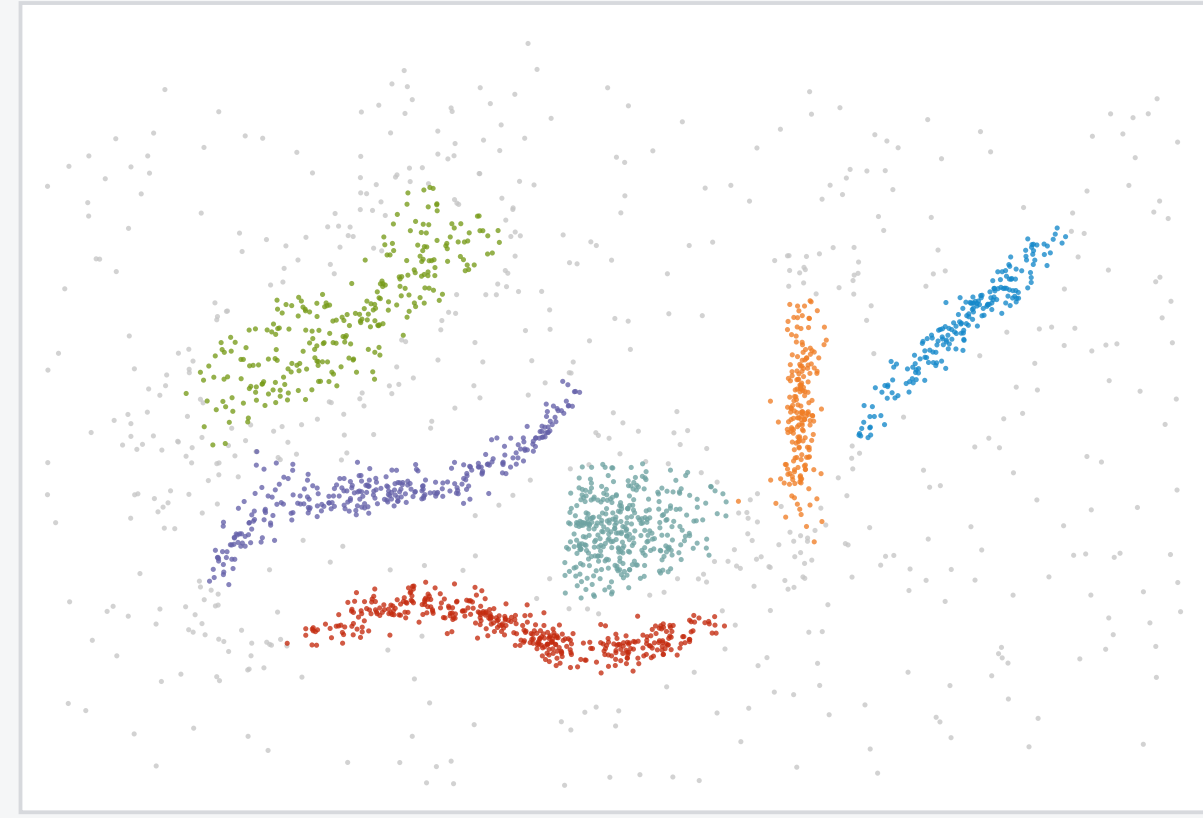
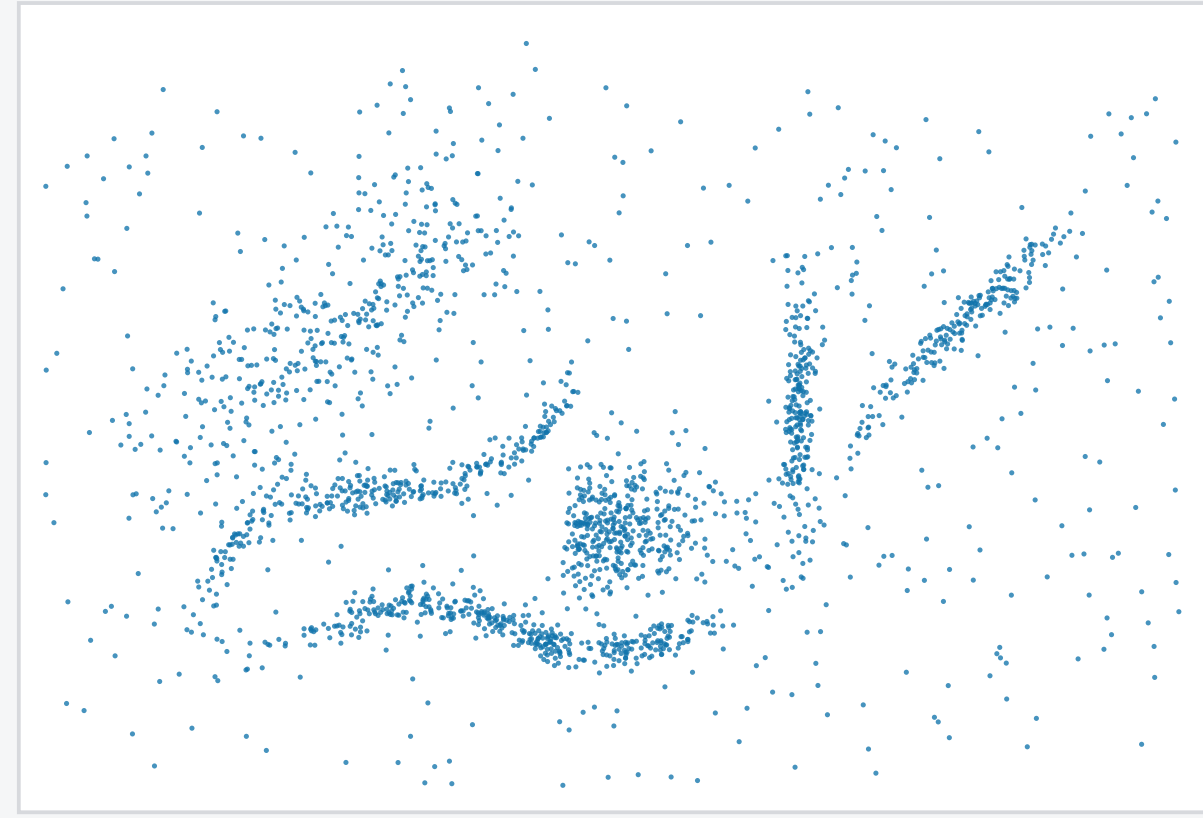
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Inria, Université Côte d'Azur · NEO & AYANA teams · 3IA-DS4H

1 Clustering

Find groups without class labels. Recover non-convex shapes and leave noisy points unassigned.

Unlabelled points

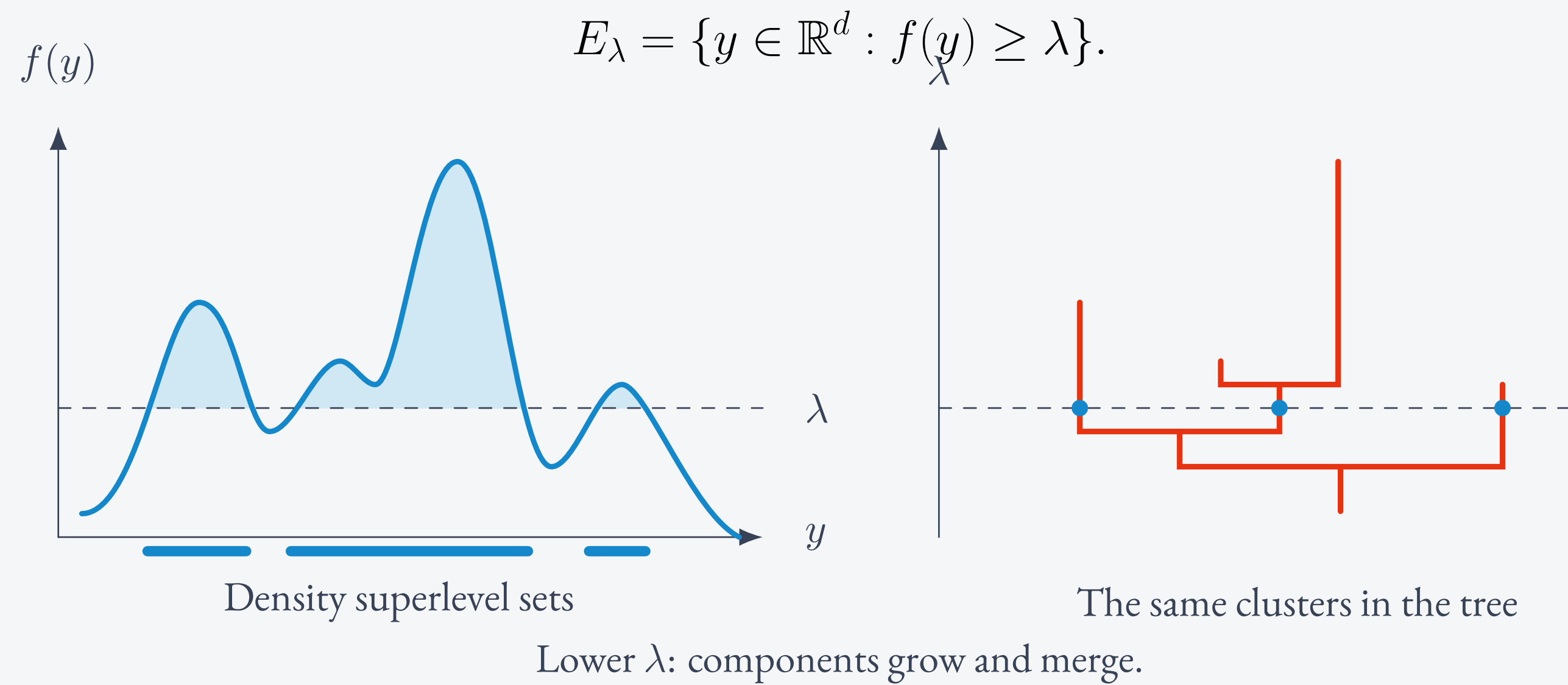
HDBSCAN [3]



Colours: clusters; grey: unassigned points [3].

2 Hartigan's density model

Clusters are connected components of density superlevel sets [1]:



K -nearest-neighbour density estimate [2]:

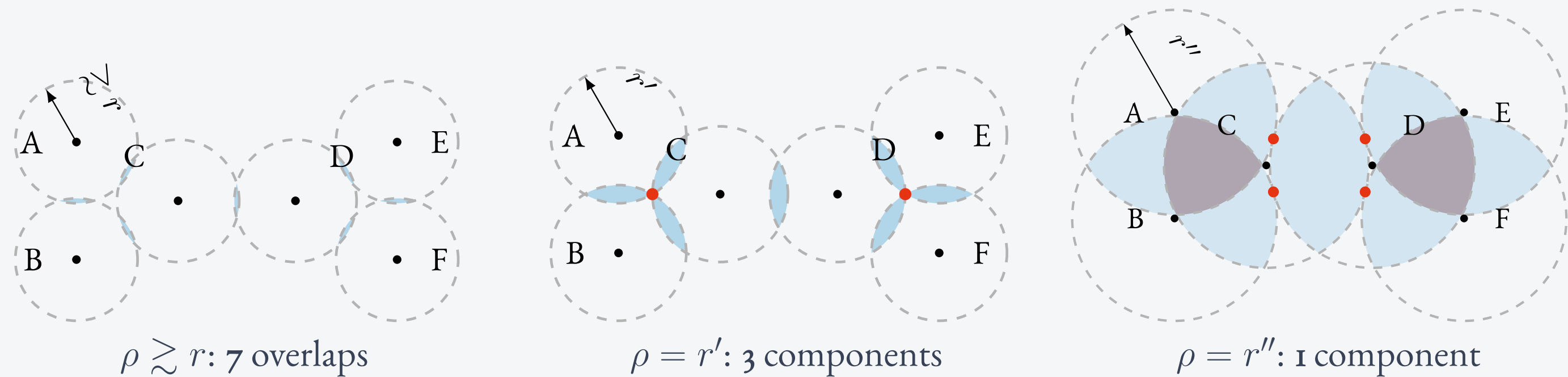
$$\hat{f}_K(y) = \frac{K}{n \omega_d r_K(y)^d}.$$

$r_K(y)$: distance to the K -th sample point; ω_d : unit-ball volume. High density means small r_K .

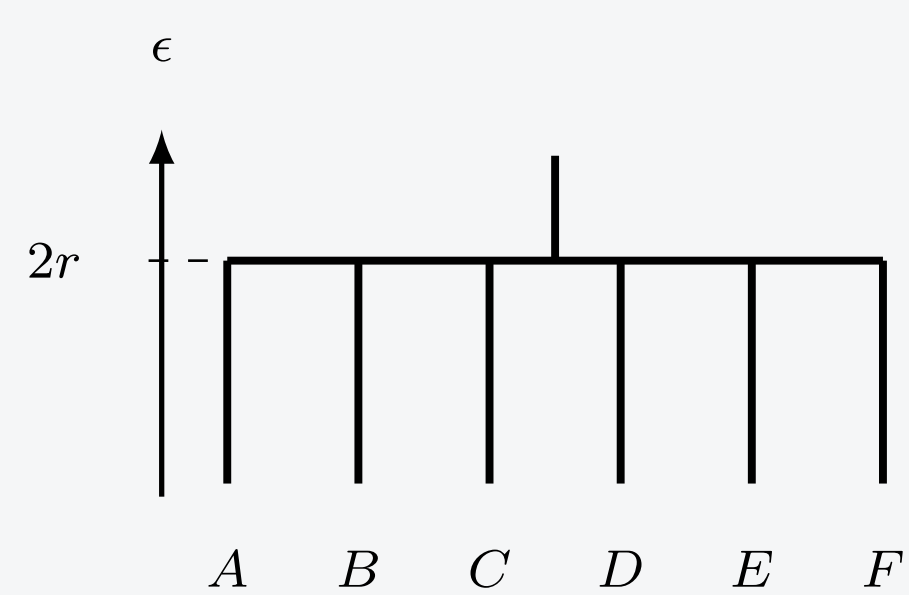
3 Same six points, different hierarchies

For $K = 2$, track the region covered by at least two balls:

$$E_{2,\rho} = \{y : r_2(y) \leq \rho\} = \bigcup_{\{x_i, x_j\} \subset X} (B(x_i, \rho) \cap B(x_j, \rho)).$$

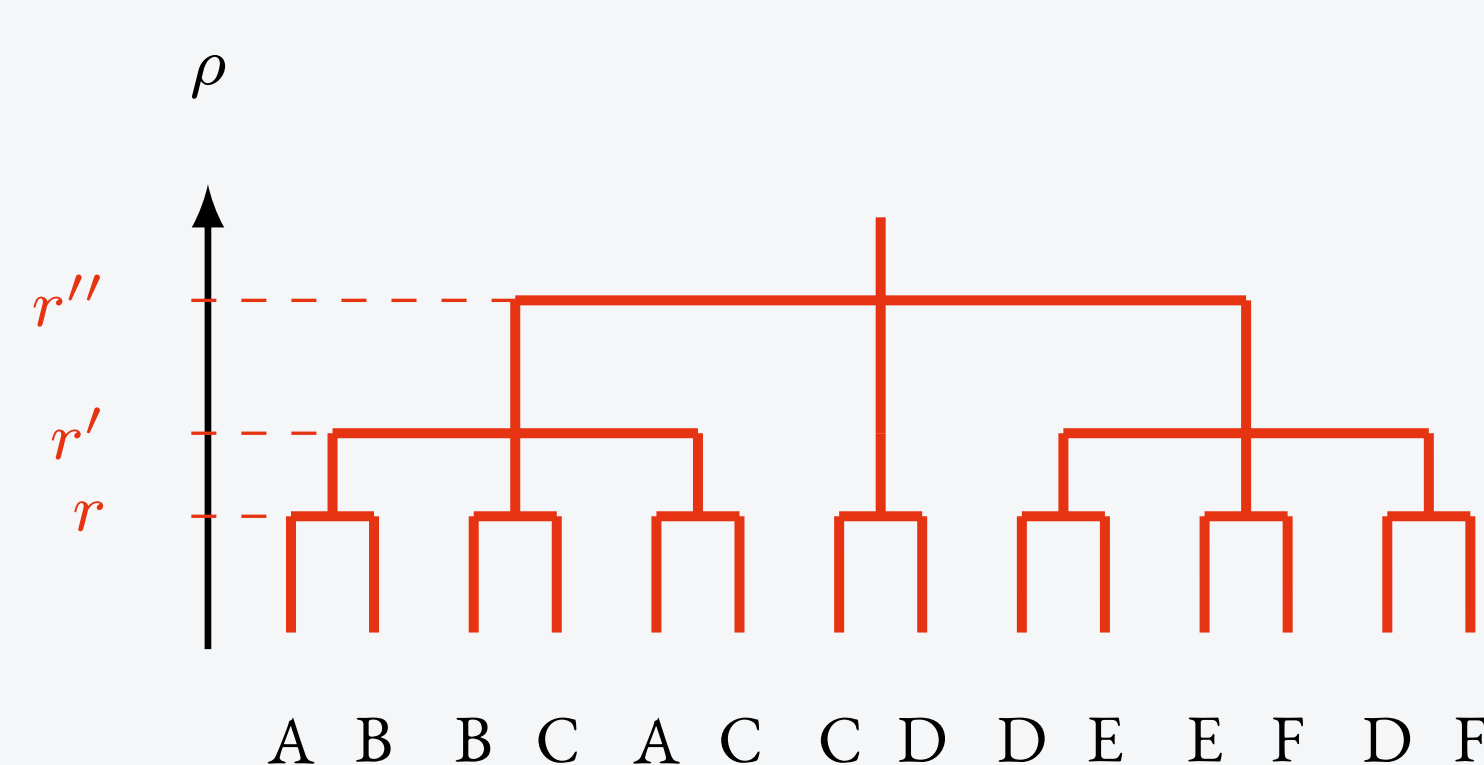


DBSCAN / Robust SL
 $K = 2$, neighbourhood radius ϵ



All six points merge at once.

2-NN HARTIGAN
Connected components of $E_{2,\rho}$



ABC, CD, DEF remain distinct until r'' .

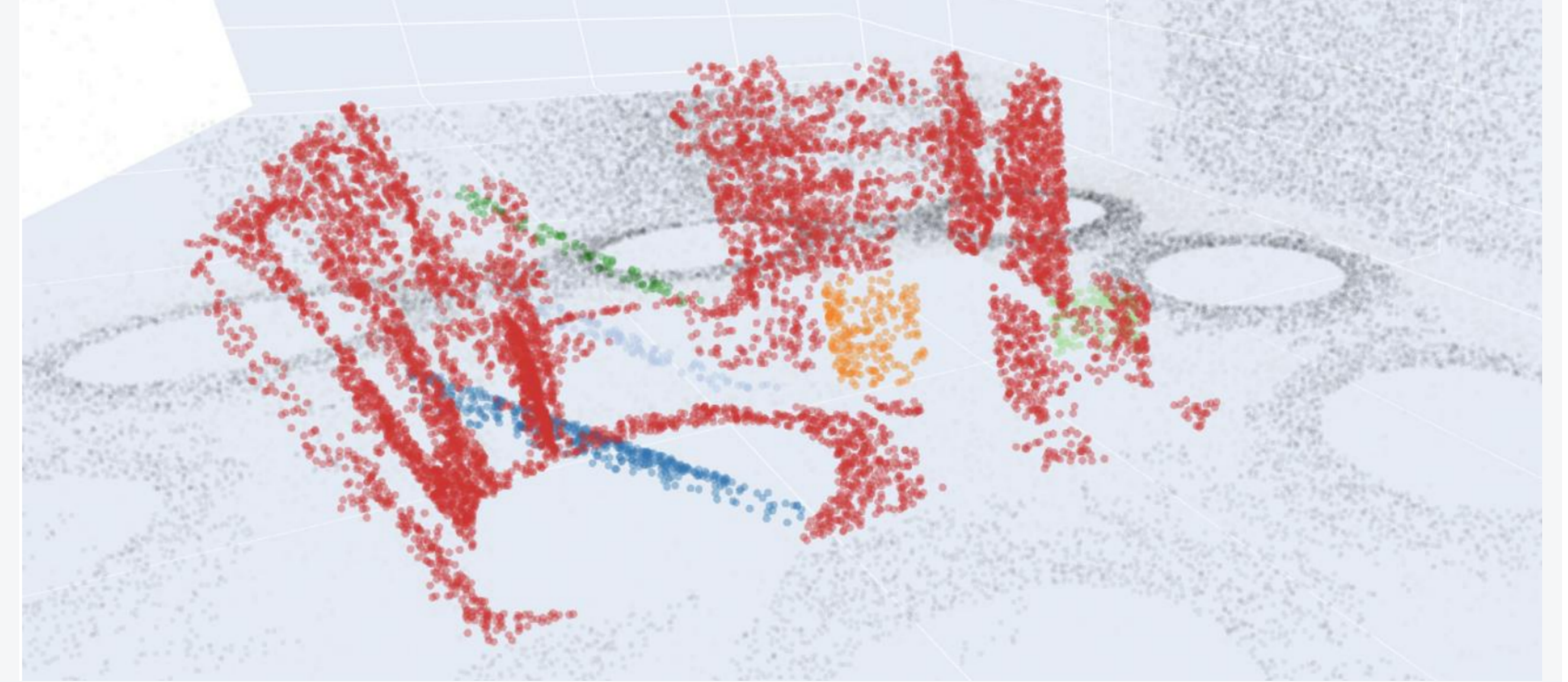
HGP-Clusterer connects edges through triangles, recovering the exact hierarchy [4,5]. Its branches may share points.

6 References

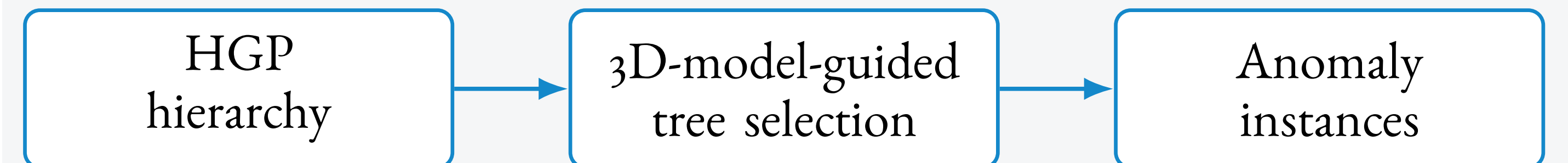
- [1] J. A. Hartigan. *Consistency of single linkage for high-density clusters*. JASA, 76:388–394, 1981.
- [2] G. Biau and L. Devroye. *Lectures on the Nearest Neighbor Method*. Springer, 2015.
- [3] R. J. G. B. Campello, D. Moulavi and J. Sander. *Density-Based Clustering Based on Hierarchical Density Estimates*. PAKDD, pp. 160–172, 2013.

4 LiDAR: detect unexpected objects

Objects present in a scan, absent from a reference 3D model. With Marie ASPRO (Inria Startup Studio).



Synthetic benchmark: 2 million points. Red: model-consistent; other colours: anomalies; grey: background. © Naval Group [6].



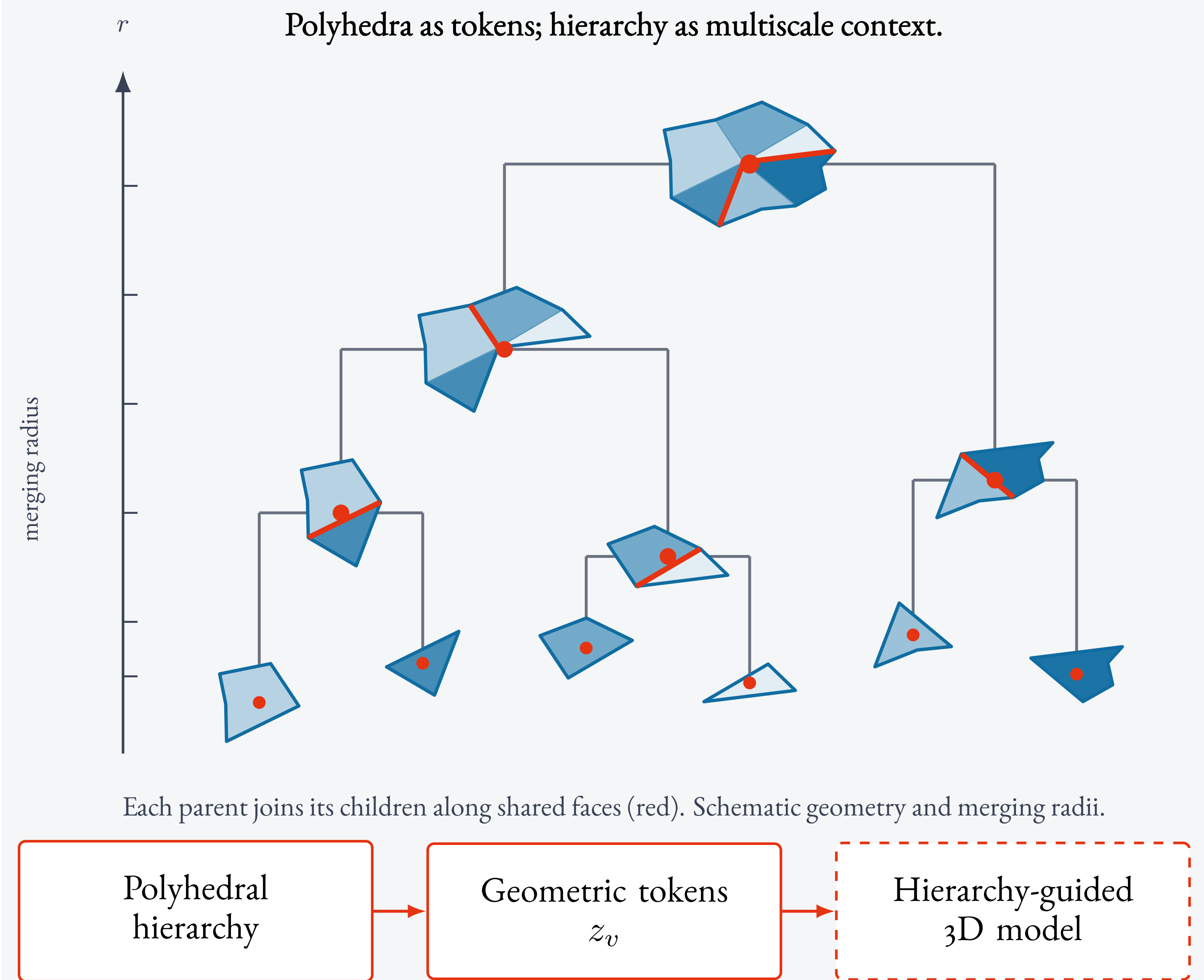
Geometric priors select branches. No training required.

5 Perspective: geometry-guided 3D learning

The same object, different sampling.



Range, scan pattern and occlusion change the point cloud. Surface geometry may be more stable.



Research hypothesis: guide a 3D foundation model with geometric tokens and their merging scales. Learning gains remain to be tested.



Percolia Louis HAUSEUX & Alban HAUSEUX
Entering Inria Startup Studio · Geometry-aware 3D perception



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